

Part IV — Rotation, Equilibrium, and Materials · Chapter 22

Stress, Strain, and Elasticity

How materials respond to forces internally — stretching, compressing, bending, and failing

Pull on both ends of a rubber band and it stretches. Release it and it snaps back. Pull far enough and it breaks. Squeeze a lump of clay and it deforms permanently. Hang a mass from a steel cable and it stretches imperceptibly — but it does stretch. Drop a ceramic plate and it shatters in an instant. These behaviors — stretching, snapping back, permanent deformation, fracture — are all responses of real materials to real forces. Understanding why different materials behave so differently is not just scientific curiosity: it is the foundation of engineering. Every bridge, building, aircraft, and medical implant depends on choosing materials that deform and fail in predictable, controlled ways.

22.1 Why Materials Deform

In the earlier chapters of this book, we treated objects as perfectly rigid — they maintained their shape no matter what forces were applied. This was a useful simplification for analyzing motion, equilibrium, and rotation. But real objects are not rigid. Every solid material deforms when forces act on it. The deformation is sometimes so small as to be invisible (a steel column under building load), sometimes obvious (a stretched spring), and sometimes permanent (a crushed car body).

Deformation occurs because materials are made of atoms and molecules held together by interatomic bonds. These bonds behave somewhat like tiny springs: they resist changes in the distances between atoms. When a force is applied to a material, it is transmitted through the network of bonds. If the force stretches the bonds, the material elongates. If it compresses them, the material shortens. If it shears them — slides one layer past another — the material distorts in shape. If the force is large enough to break bonds entirely, the material fractures.

The key insight is that forces applied at the surface of a material create internal forces throughout the material. These internal forces — the forces that atoms exert on one another as they resist deformation — are the subject of this chapter. Understanding them allows us to predict how much a structure will deform under load, whether it will spring back or permanently change shape, and at what load it will ultimately fail.

22.2 Internal Forces and Material Response

When you pull on the ends of a rod, the applied forces must be balanced by internal forces throughout the material. Imagine slicing the rod with an imaginary plane perpendicular to its length. The left portion exerts a force on the right portion across that plane, and the right exerts an equal and opposite force on the left. These are the internal forces that resist the applied load.

The magnitude of these internal forces depends on the applied load. The distribution of these forces across a cross-section determines whether the material will hold or fail. If the internal forces per unit area exceed the material's breaking threshold, bonds rupture and the material fails.

There are three fundamental types of internal force (and corresponding deformation): tension (pulling apart), compression (pushing together), and shear (sliding one layer past another). Bending combines tension and compression simultaneously — one side of a bent beam is in tension and the other in compression. Torsion (twisting) involves shear throughout the material.

Most engineering analyses of structures begin by identifying which type of internal force dominates in each member, then applying the appropriate material properties to calculate whether the member can withstand the load.

22.3 Stress

The first key concept for describing how a material responds to load is stress. Stress (symbol: sigma, σ) is the internal force per unit cross-sectional area. It describes how intensely the material is being loaded — how much force is being transmitted through each unit of cross-section.

$$\text{Stress: } \sigma = F / A$$

where F is the applied force (N) and A is the cross-sectional area (m^2)

Unit of stress: Pascal (Pa) = N/m^2

The reason stress (force per area) is more useful than raw force alone is that it allows comparison between different sizes of the same material. A 1 mm diameter wire and a 10 mm diameter wire are made of the same steel. Apply 100 N to each. The thin wire experiences stress of $100 \text{ N} / (\pi \times (0.0005)^2) \approx 127 \text{ MPa}$. The thick wire experiences stress of $100 \text{ N} / (\pi \times (0.005)^2) \approx 1.27 \text{ MPa}$. The thin wire is under 100 times greater stress — far closer to its failure limit — even though the applied force is identical. Stress correctly captures this difference; force alone does not.

Tensile stress occurs when the material is being pulled (stretched). Compressive stress occurs when the material is being pushed (compressed). Shear stress occurs when the internal force acts parallel to the cross-section, tending to slide one part past another.

Stress is measured in pascals (Pa), where $1 \text{ Pa} = 1 \text{ N}/\text{m}^2$. Because material stresses in engineering are typically very large, practical units are megapascals ($\text{MPa} = 10^6 \text{ Pa}$) and gigapascals ($\text{GPa} = 10^9 \text{ Pa}$). Steel has a typical tensile strength of around 400–700 MPa.

22.4 Strain

Stress describes how hard the material is being pushed or pulled. Strain describes how much it has actually deformed. Strain (symbol: epsilon, ϵ) is the fractional change in a dimension — the ratio of the change in length to the original length.

$$\text{Strain: } \epsilon = \Delta L / L_0$$

where ΔL is the change in length and L_0 is the original length

Strain is dimensionless (no units — it is a ratio of two lengths)

Because strain is a ratio, it is dimensionless. A strain of 0.01 means the material has stretched (or compressed) by 1% of its original length. A strain of 0.001 (0.1%) is typical for steel under normal engineering loads. Rubber can sustain strains of several hundred percent before breaking.

Strains in engineering structures are typically very small — often less than 0.001. A steel bridge girder under traffic load might strain by only 0.0005 (0.05%), stretching 0.5 mm for every meter of length. Although this seems negligible, it determines whether the structure is safe, and measuring such tiny strains precisely is the work of experimental stress analysis.

Like stress, strain comes in different types corresponding to the mode of deformation. Tensile strain corresponds to elongation under tensile stress. Compressive strain corresponds to shortening under compressive stress. Shear strain describes the angular distortion of a shape under shear stress.

22.5 Elastic vs. Plastic Deformation

Materials deform in two fundamentally different ways depending on whether the applied stress exceeds a critical threshold called the yield strength.

Elastic deformation is reversible. When the load is removed, the material returns to its original shape and dimensions. The interatomic bonds have been stretched or compressed but not broken or permanently rearranged. A rubber band stretched slightly, a spring compressed, or a steel beam deflected under a load — all exhibit elastic deformation as long as they remain within their elastic limit. This is the normal operating regime for most engineering structures: everything should deform elastically and recover fully when the load is removed.

Plastic deformation is permanent. When stress exceeds the yield strength, atoms slip past one another into new equilibrium positions. When the load is removed, the material does not return

to its original shape — it has been permanently distorted. A bent paper clip, a hammered nail, a crushed car body: all have undergone plastic deformation. The material is not broken, but it has permanently changed shape.

The boundary between elastic and plastic behavior is the yield point (or elastic limit). Below this stress, deformation is elastic and reversible. Above it, deformation becomes partly or fully plastic and permanent. Understanding this boundary is critical for engineering: structures must be designed to operate entirely within the elastic regime under normal loads, with sufficient margin so that extraordinary loads do not cause permanent deformation that could compromise the structure.

22.6 Hooke's Law Revisited in Materials

You first encountered Hooke's law in Chapter 9, where it described the force-extension relationship for springs: $F = k \cdot x$. Hooke's law applies to springs because springs are made of materials that exhibit elastic behavior — and the law applies to those materials directly.

In the context of materials science, Hooke's law states that stress is proportional to strain in the elastic regime: stress = $E \times$ strain ($\sigma = E \cdot \epsilon$), where E is the elastic modulus of the material. This is the material-level version of the spring law. The spring constant k from Chapter 9 is related to both the material property E and the geometry (length and cross-section) of the spring.

Hooke's Law for materials (elastic region):

$$\sigma = E \cdot \epsilon$$

This linear relationship holds until the yield point is reached.

The proportionality between stress and strain in the elastic regime is why elastic deformation is fully recoverable: the atoms are displaced from their equilibrium positions in a way that generates a restoring force proportional to the displacement — exactly analogous to a spring. Remove the load, and the restoring forces bring all atoms back to their original positions.

This proportionality breaks down at the yield point. Beyond this point, atomic slip occurs: atoms rearrange permanently, the stress-strain curve is no longer linear, and the deformation is no longer recoverable.

22.7 Young's Modulus (Conceptual Introduction)

Young's modulus (symbol: E) is the fundamental measure of a material's stiffness — its

resistance to elastic deformation under tensile or compressive stress. It is defined as the ratio of tensile (or compressive) stress to tensile (or compressive) strain in the elastic region: $E = \sigma / \epsilon$.

A material with a high Young's modulus is stiff: it requires a large stress to produce a small strain. Diamond has the highest Young's modulus of any natural material (about 1000 GPa) — it is essentially incompressible under normal stresses. Steel has $E \approx 200$ GPa; a large force is needed to stretch it even slightly. Rubber has E in the range of 0.01–0.1 GPa — it stretches enormously under small stress.

Young's modulus is a property of the material, not the object. A thick steel rod and a thin steel wire have the same Young's modulus but very different spring constants: the rod is much stiffer because its larger cross-section reduces the stress for a given force, and its shorter length reduces the extension for a given strain.

Material	Young's Modulus (GPa)	Typical Strength (MPa)	Character
Diamond	~1000	~10,000 (compressive)	Very stiff, brittle
Steel	~200	400–700	Stiff, ductile
Aluminium	~70	200–400	Moderately stiff, ductile
Glass	~65	~50 (tensile)	Stiff, brittle
Concrete	~30	~30 (compressive)	Stiff, very brittle
Wood (along grain)	~10	~40 (compressive)	Moderately stiff, varies
Rubber	0.01–0.1	~15	Very flexible, tough

Table 22.1 Approximate Young's moduli and tensile strengths for common materials. Values vary significantly with composition, temperature, and processing.

Young's modulus is one of three fundamental elastic moduli — the others are the shear modulus (G , relating shear stress to shear strain) and the bulk modulus (K , relating pressure to volumetric compression). At this level we focus on Young's modulus, which is most directly relevant to everyday engineering situations involving elongation and compression.

Bridge Note *University materials science and structural engineering courses use all three moduli, along with Poisson's ratio, to describe the full three-dimensional elastic behavior of materials. The concepts introduced here — stress, strain, and Young's modulus — are the mandatory first step.*

22.8 Stress–Strain Curves

A stress–strain curve is a graph of stress (vertical axis) versus strain (horizontal axis) obtained by stretching a sample of material at a controlled, slow rate while measuring the force and extension continuously. It is the most informative single description of a material's mechanical behavior.

For a ductile metal like mild steel, the curve has a characteristic shape with distinct regions:

- Elastic region: The initial straight-line portion where stress and strain are proportional (Hooke's law holds). The slope of this line is Young's modulus. The material deforms and fully recovers when the load is removed.
- Yield point: The stress at which the curve deviates from linearity. Beyond this point, plastic deformation begins. For steel, there is often a distinct upper and lower yield point — a sudden 'drop' in the curve as yielding begins.
- Plastic region (strain hardening): After yielding, many metals can sustain increasing stress as plastic deformation proceeds. This is strain hardening — the material actually becomes stronger as dislocations (defects in the crystal structure) pile up and impede further slip.
- Ultimate tensile strength: The maximum stress the material can sustain. This is the highest point on the curve. Beyond this, the sample begins to neck (thin locally), concentrating stress in the narrowed region.
- Fracture point: The stress at which the sample breaks. For ductile materials this is usually lower than the ultimate tensile strength because necking has reduced the actual cross-section.

The area under the stress-strain curve has physical significance: it represents the energy per unit volume absorbed by the material up to the point of fracture. This is related to the material's toughness — its ability to absorb energy without breaking. A material with a large area under its stress-strain curve is tough even if it is not extremely strong.

Different materials produce dramatically different stress-strain curves. A brittle material like glass or ceramic has a short, steep elastic region followed by sudden fracture — no plastic region at all. A rubber-like material has a long, low, non-linear elastic region. Knowing the shape of the stress-strain curve tells an engineer at a glance whether a material is stiff, flexible, brittle, ductile, or tough.

22.9 Yield Strength and Failure

The yield strength is the stress at which a material transitions from elastic to plastic behavior — the stress at the yield point on the stress-strain curve. It is one of the most important design parameters in engineering, because it marks the upper limit of the material's safe working range for structures that must not permanently deform.

Engineering components are typically designed to operate at stresses well below the yield strength, incorporating a safety factor. A safety factor of 2, for example, means the component is designed to withstand twice the maximum expected load before yielding. Safety factors

account for uncertainties in the load, material variability, manufacturing defects, and degradation over time.

The ultimate tensile strength (UTS) is the maximum stress the material can withstand before fracture. For ductile materials, the UTS exceeds the yield strength (strain hardening allows the material to carry more stress after yielding). For brittle materials, the UTS and fracture stress are effectively the same — fracture occurs with no plastic deformation.

Fatigue failure is a particularly important mode of failure not captured by static stress analysis. A material subjected to repeated cycles of stress — even well below the yield strength — can fail after many thousands or millions of cycles. Microscopic cracks initiate at stress concentrations and grow slowly with each cycle until the remaining cross-section is too small to carry the load, and sudden fracture occurs. Fatigue accounts for the majority of service failures in metal structures. Aircraft, engine components, and bridges all require careful fatigue analysis.

22.10 Brittleness, Ductility, and Toughness

Three terms describe a material's overall behavior under load beyond the elastic range: ductility, brittleness, and toughness. These are related but distinct concepts, and confusing them is a common source of error.

Ductility is the ability of a material to undergo large plastic deformation before fracturing. A ductile material can be drawn into wires (ductility originally referred specifically to wire-drawing), bent, hammered, or stretched significantly without breaking. Copper, gold, aluminium, and low-carbon steel are highly ductile. Ductility is valuable in structural applications because ductile materials give warning before failure — they visibly deform, deflect, and make noise as they approach their limits, giving occupants of a building time to evacuate and engineers time to notice and respond.

Brittleness is the tendency to fracture with little or no plastic deformation. A brittle material fails suddenly, with no warning, when stress exceeds its fracture strength. Glass, ceramics, cast iron, and concrete under tension are brittle. Brittle failure is catastrophic: the structure goes from fully intact to completely failed with no intermediate deformation. This is why brittle failure in structures is far more dangerous than ductile failure.

Toughness is the ability of a material to absorb energy before fracturing. It depends on both strength and ductility — a tough material must be strong enough to resist high stress and ductile enough to deform significantly before fracturing. Graphically, toughness corresponds to the area under the stress-strain curve. A very strong but brittle material (like hardened steel) may have less toughness than a moderately strong but very ductile material (like soft copper), because the brittle material fractures before absorbing much energy.

Property	Definition	High Example	Low Example
Stiffness	Resistance to elastic	Diamond, steel	Rubber, foam

	deformation ($E = \sigma/\epsilon$)		
Strength	Maximum stress before yield or fracture	Tool steel, titanium	Lead, soft clay
Ductility	Capacity for plastic deformation before fracture	Copper, gold, mild steel	Glass, ceramics, cast iron
Brittleness	Tendency to fracture without plastic deformation	Glass, concrete, cast iron	Mild steel, rubber
Toughness	Energy absorbed before fracture (area under σ - ϵ curve)	Structural steel, Kevlar	Glass, hardened tool steel
Hardness	Resistance to surface indentation or scratching	Diamond, hardened steel	Lead, gold

Table 22.2 Mechanical properties of materials — definitions and representative examples.

It is important not to conflate strength and toughness. A material can be strong and brittle (glass, hardened steel) — it holds a lot of stress but shatters without warning. A material can be weak and tough (soft rubber) — it yields at low stress but absorbs a great deal of energy before tearing. Optimal material selection depends on which combination of properties the application demands.

22.11 Material Selection in Engineering

Engineers select materials for a component based on the mechanical, thermal, chemical, and economic requirements of the application. Understanding stress-strain behavior is the starting point, but the selection process is rarely simple because different requirements often conflict.

Structural steel is the dominant material for large-scale structures — buildings, bridges, ships — because it is strong, ductile (giving warning of overload), relatively inexpensive, available in large quantities, and well understood by engineers. Its main weaknesses are susceptibility to corrosion and relatively high density.

Aluminium alloys are used when low weight is important alongside strength — aircraft frames, bicycle frames, automotive components. Aluminium is about one-third the density of steel but has roughly half the stiffness and strength, so structures must be designed differently. Aluminium also does not rust, which is valuable in marine environments.

Concrete is excellent in compression but extremely weak in tension — it fractures at tensile stresses of only about 3–5 MPa. This is why concrete structures are almost always reinforced with steel: the concrete handles compressive loads and the embedded steel bars handle tensile loads. Prestressed concrete goes further: the steel is pre-tensioned so that the concrete is

always in compression even when the beam deflects under a load.

Carbon fiber reinforced polymer (CFRP) has become important in high-performance applications — aerospace, racing cars, sports equipment — because it offers extremely high strength and stiffness per unit weight. It is far more expensive than steel and difficult to manufacture in complex shapes, but where weight is critical and cost is secondary, it has no peer.

Biological materials such as bone, tendon, wood, and shell exploit composite structures to achieve properties not available in single-component materials. Bone, for example, combines a stiff ceramic phase (hydroxyapatite) with a tough protein phase (collagen) to produce a material that is both stiff and fracture-resistant — a combination that neither component achieves alone. Understanding and mimicking biological materials is one of the most active areas of modern materials science.

22.12 Common Mistakes in Material Analysis

Students and beginning engineers frequently make predictable errors when applying stress and strain concepts. Recognizing these mistakes is the first step to avoiding them.

- Confusing stress with force. Stress is force per unit area. A small force concentrated on a tiny area produces enormous stress. A large force distributed over a huge area produces small stress. Always ask: what is the cross-sectional area through which the force is transmitted?
- Confusing strain with deformation. Strain is fractional deformation ($\Delta L/L_0$), not absolute deformation. A 1 mm extension of a 10 m cable is a strain of 0.0001 (0.01%). A 1 mm extension of a 1 cm spring is a strain of 0.1 (10%). Same deformation, radically different strains.
- Confusing stiffness (Young's modulus) with strength (yield or ultimate stress). A stiff material is not necessarily strong. Glass is stiff ($E \approx 65$ GPa) but weak in tension (fails at ~ 50 MPa). Steel is also stiff ($E \approx 200$ GPa) and strong (~ 500 MPa). Rubber is not stiff but is tough.
- Assuming brittle materials are weak. Glass and ceramics can sustain very high compressive stresses. They fail at low tensile stress. Concrete compressive strength (30 MPa) exceeds its tensile strength (3 MPa) by a factor of 10.
- Applying Young's modulus beyond the elastic region. The relationship $\sigma = E \cdot \epsilon$ only holds in the elastic region, up to the yield point. Do not use it to calculate stress in the plastic region.
- Ignoring the cross-sectional area when applying the stress formula. If a bar with a non-uniform cross-section is under load, different cross-sections experience different stresses. The narrowest section is always at greatest risk because it has the smallest area for the same force.

Chapter Summary

Materials deform when forces act on them because interatomic bonds resist changes in separation distance. External forces create internal forces distributed throughout the material.

Stress ($\sigma = F/A$) is the internal force per unit cross-sectional area. It is measured in pascals (Pa). Stress — not force alone — determines whether a material will yield or fracture, because it accounts for how the force is distributed over area.

Strain ($\epsilon = \Delta L/L_0$) is the fractional change in length. It is dimensionless. Strain — not extension alone — characterizes how much a material has deformed relative to its original size.

Elastic deformation is reversible: the material returns to its original shape when the load is removed. Plastic deformation is permanent: atoms have slipped into new positions and the material does not recover.

In the elastic regime, stress is proportional to strain: $\sigma = E \cdot \epsilon$. Young's modulus E is the constant of proportionality and measures material stiffness. High E means the material is stiff; low E means it is flexible.

The stress–strain curve graphically describes a material's complete mechanical behavior. Key points are the yield strength (onset of plasticity), ultimate tensile strength (maximum stress), and fracture point.

Ductile materials deform plastically before fracturing (giving warning). Brittle materials fracture suddenly with no plastic deformation (dangerous in structures). Toughness is the energy absorbed before fracture — the area under the stress-strain curve.

Material selection requires matching mechanical properties (stiffness, strength, ductility, toughness) to application requirements (load type, weight, cost, environment). Steel, aluminium, concrete, CFRP, and biological composites each excel in specific roles.

Key Terms

Term	Definition
Stress (σ)	Internal force per unit cross-sectional area; $\sigma = F/A$; measured in pascals (Pa = N/m ²).
Strain (ϵ)	Fractional change in length under load; $\epsilon = \Delta L/L_0$; dimensionless.
Tensile stress/strain	Stress/strain due to pulling (stretching) forces.
Compressive stress/strain	Stress/strain due to pushing (compression) forces.

Shear stress/strain	Stress/strain due to forces acting parallel to a cross-section, tending to slide layers past one another.
Elastic deformation	Reversible deformation; material returns to original shape when load is removed; occurs below the yield point.
Plastic deformation	Permanent deformation; atoms are rearranged and the material does not recover when load is removed; occurs above the yield point.
Young's modulus (E)	The ratio of tensile (or compressive) stress to tensile (or compressive) strain in the elastic region; the measure of material stiffness; $E = \sigma/\epsilon$.
Yield strength	The stress at which a material transitions from elastic to plastic behavior; the upper limit of safe operating stress for structures that must not deform permanently.
Ultimate tensile strength	The maximum stress a material can sustain before fracturing; the peak of the stress-strain curve.
Stress-strain curve	A graph of stress versus strain obtained from a tensile test; characterizes a material's complete mechanical response.
Ductility	The ability of a material to undergo significant plastic deformation before fracturing.
Brittleness	The tendency to fracture with little or no plastic deformation.
Toughness	The energy absorbed by a material per unit volume before fracture; represented by the area under the stress-strain curve.
Safety factor	The ratio of the material's yield (or ultimate) strength to the maximum designed working stress; accounts for uncertainty in loads and material properties.

Concept Review Questions

Answer the following in complete sentences. No calculations required unless stated.

1. Explain why stress (not force) determines whether a material yields. Give an example that shows two objects with the same applied force can experience very different stresses.
2. A steel wire and an aluminium wire of identical dimensions carry identical loads. Which experiences greater stress? Which experiences greater strain? Explain using Young's modulus.
3. Explain the difference between elastic and plastic deformation using the concept of interatomic bonds. Why is elastic deformation reversible but plastic deformation permanent?

4. Sketch a qualitative stress-strain curve for (a) a ductile metal like steel, and (b) a brittle material like glass. Label the key features of each.
5. A material is described as 'very strong but not very tough.' What does this mean? Give a material example and explain why this combination of properties might be problematic in a structural application.
6. Explain why concrete in structural applications is almost always reinforced with steel. What does each material contribute to the composite?
7. What is a safety factor? Explain why safety factors are necessary even when the expected maximum load is known precisely.
8. Why is brittle failure in a structure more dangerous than ductile failure? Think about the engineering consequences of each mode.

Stress and Strain Problems

9. A steel bolt has a circular cross-section with diameter 8.0 mm. A tensile load of 5000 N is applied along its axis. Calculate the tensile stress in the bolt (in MPa).
10. A 3.0 m aluminium rod stretches by 0.90 mm when a tensile load is applied. Calculate the tensile strain.
11. A cable 10.0 m long stretches by 2.0 mm under load. Calculate the strain. If the cable is made of a material with Young's modulus 150 GPa, what is the stress?
12. A concrete column has a cross-sectional area of 0.04 m². It supports a compressive load of 800 kN. Calculate the compressive stress. If the column's compressive strength is 30 MPa, what is the safety factor?
13. A steel wire ($E = 200$ GPa) of length 2.0 m and cross-sectional area 1.5×10^{-6} m² is subjected to a tensile stress of 120 MPa. Calculate (a) the strain and (b) the extension of the wire.
14. Two wires of identical material ($E = 80$ GPa) carry the same load. Wire A has twice the length and twice the cross-sectional area of Wire B. Compare their stresses, strains, and extensions.

Elasticity Exercises

15. A spring has a spring constant of 500 N/m. It is made of steel wire with cross-section 0.5 mm² and total wire length 0.40 m. Estimate the maximum stress in the wire when the spring is compressed by 30 mm. Is this within the elastic range of steel (yield strength ~250 MPa)?
16. A rubber band ($E \approx 0.02$ GPa) has a cross-sectional area of 2.0 mm² and a natural length of 80 mm. It is stretched to 120 mm. Calculate the strain, the stress, and the force required to maintain this extension.

17. Hooke's law for springs states $F = kx$. Show that, for a uniform cylindrical rod of Young's modulus E , cross-sectional area A , and length L , the spring constant is $k = EA/L$. Use this to explain why a rod's stiffness increases with area and decreases with length.

Material Comparison Analysis

Use the data in Table 22.1 and your knowledge of material properties to answer the following.

18. A structural member must resist 200 MPa of tensile stress. Which materials from Table 22.1 could be used? Which would be unsuitable? Explain your reasoning.
19. Two candidates for a lightweight aircraft bracket are aluminium ($E = 70$ GPa, density = 2700 kg/m^3) and steel ($E = 200$ GPa, density = 7800 kg/m^3). Compare the specific modulus ($E/\text{density}$) of each. Which provides greater stiffness per unit mass?
20. A dam must resist very high compressive stresses from water pressure. A builder proposes using concrete. Evaluate this choice: what are the advantages and the critical limitation? What reinforcement might address the limitation?
21. Explain why glass fiber reinforced polymer (fiberglass) can be a better choice than either glass or polymer alone for a boat hull. What mechanical property does the composite achieve that neither component provides individually?

Engineering Application Problems

22. A suspension bridge has steel cables each with a cross-sectional area of 0.12 m^2 . Each cable must support a maximum tensile load of 15 MN. Calculate the maximum tensile stress. If the steel's yield strength is 500 MPa, what is the safety factor? Is this adequate for a public bridge?
23. A hip replacement implant is made of titanium alloy ($E \approx 110$ GPa, yield strength ≈ 900 MPa). During normal walking, the implant experiences a compressive load of 3000 N distributed over a cross-sectional area of 200 mm^2 . (a) Calculate the stress. (b) Calculate the strain. (c) What is the safety factor against yielding?
24. A pre-stressed concrete bridge beam is designed so that the steel tendons are pre-tensioned to 700 MPa before the concrete sets. Explain why this puts the concrete in compression, and why this improves the beam's ability to carry loads that would otherwise put the concrete in tension.
25. A designer wants to select a material for a safety clip that must bend plastically (and not fracture) when overloaded, providing a visible warning without catastrophic failure. Should they choose a high-strength brittle material or a lower-strength ductile material? Justify your answer fully using stress-strain concepts.

Chapter 22 Checkpoint

Before moving on to Part V, confirm you can do each of the following without notes.

I can...	See Section
Explain why materials deform using the concept of interatomic bonds	22.1
Identify the three types of internal force (tension, compression, shear)	22.2
Define stress and calculate it using $\sigma = F/A$; state its unit	22.3
Define strain and calculate it using $\epsilon = \Delta L/L_0$; explain why it is dimensionless	22.4
Distinguish elastic from plastic deformation and identify the yield point on a stress-strain curve	22.5
State Hooke's law for materials as $\sigma = E \cdot \epsilon$ and identify where it applies	22.6
Define Young's modulus and compare it qualitatively across materials	22.7
Sketch and label the key features of a stress-strain curve for a ductile material	22.8
Define yield strength, ultimate tensile strength, and safety factor	22.9
Distinguish ductility, brittleness, and toughness; give an example of each	22.10
Explain why material selection for an engineering application requires matching multiple properties	22.11
Identify and avoid the most common errors in stress-strain analysis	22.12

End of Part IV: Rotation, Equilibrium, and Materials

Continue to Part V →